Homework 8: Multiple regression model

Carlo Gaetan (but see the credits)

Contents

Credits: Material from a course taught by Mark Scheuerell.

Background

Your task is to analyze some data on the concentration of nitrogen in the soil at 41 locations on the island of Maui in the Hawaiian Archipelago. Along with the nitrogen measurements, there are 4 possible predictor variables that may help to explain the variation in soil nitrogen concentration. The accompanying data file `soil_nitrogen.csv` has the following 5 columns of data:

- `nitrogen`: concentration of soil nitrogen (mg nitrogen kg^{-1} soil)
- `temp`: average air temperature (°C)
- `precip`: average precipitation (cm)
- `slope`: slope of the hillside (degrees)
- `aspect`: aspect of the hillside (N, S)

Problems

a) Begin by building a global model that contains all four of the predictors plus an intercept. Show the resulting summary table, and report the multiple and adjusted R^3 values. Also report the estimate of the residual variance $\hat{\sigma}^2$.

b) Comment the results about the significance of the estimated coefficients

c) Check the residuals from your full model for possible violations of the assumption that the $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$.

d) Does this seem like a reasonable model for these data? Why or why not?

e) Now fit various models using all possible combinations of the 4 predictors, including an intercept-only model (ie, there should be a total of 16 models). Compute the AIC, AICc, and BIC for each of your models and compare the relative rankings of the different models.

f) Use a stepwise selection procedure for select the "best" model

g) Use this model to predict what the soil nitrogen concentration would be on the nearby island of Moloka'i if the average precipitation was 150 cm, the average temperature was 22 °C, and the hillside faced south with a slope of 11 degrees.